a BFF contracts: detection and persistence

Variant	Worlds N	Modal worlds, n	Population worlds, n	Collapse n / pop. n
Standard	24	9	7	0/7
Literal, no wrap	12	12	5	5/5
Standard, harmless	12	7	3	0/3
Pointer wrap	24	19	19	0/19
Wrap + literal	12	12	12	12/12
Wrap + literal, harmless	12	12	12	0/12

Harmless: unmatched brackets do not halt; literal emits operands

BFF = Brainfuck-family machine; wrap permits pointer wrap

Modal: generation-2 parent-byte-match gain ≥0.3 (headroom-normalized)
Population: ≥50% cells heritable (same gain ≥0.3 criterion)
Later collapse: terminal heritable-cell fraction <10% after population detection

b First-modal tapes: fetch / inflow counts

Fetch/256	H (bits)	Tapes, n
Standard (n=9)		
0.00000	0.00000	4
0.00000	7.24969	1
0.00391	0.03687	2
0.00391	2.06372	1
0.19531	7.39789	1
Literal, no wrap (n=12)		
1.00000	7.97656	1
1.00000	7.98143	2
1.00000	7.98438	2
1.00000	7.99219	2
1.00000	8.00000	5
Wrap + literal, harmless (n=12)		
1.00000	0.00000	4
1.00000	0.03687	6
1.00000	0.07373	1
1.00000	0.11056	1

Fetch/256	H (bits)	Tapes, n
Pointer wrap (n=19)		
0.00000	0.00000	11
0.00000	0.03687	2
0.00000	1.92163	1
0.00000	2.52233	1
0.00000	7.07753	1
0.00000	7.16823	1
0.00781	0.07373	1
0.00781	1.73558	1
Wrap + literal (n=12)		
1.00000	0.58729	3
1.00000	0.62380	1
1.00000	0.69675	1
1.00000	0.73319	1
1.00000	0.76960	2
1.00000	0.80599	2
1.00000	0.84236	1
1.00000	0.98759	1

64 first-modal tapes, one/world; n counts each fetch/H pair
Fetch: encounters with any partner fetch /256; values rounded to 5 decimals
H = H(offspring | fixed parent), bits; 256 uniform partners
Standard, harmless: paired fetch/H archive unavailable

c Bandwidth controls: open fetch, zero inflow

Operand copies r	Worlds	H bits	Fetch/256
1	12	8.0	1.00
2	6	0.0	1.00
3	6	0.0	1.00

Background: Literal, no wrap; vary operand copies r

Literal P a b emits a,b repeated r times

d Halting dial: all worlds included in counts

Halt p	Worlds N	P persistence worlds, n/N	Median final P share all worlds
0	12	12/12	0.984
0.01	6	6/6	0.984
0.03	6	6/6	0.984
0.1	6	6/6	0.984
0.3	6	5/6	0.102
1	12	0/12	0.002

Background: Wrap + literal; p = 0 harmless, p = 1 halting

World medians: conditional H and any-partner-fetch encounters/256
r≥2: zero partner inflow with fetch in every encounter

p = probability an unmatched bracket halts; P = literal-copy instruction
Persistence: P-tape share ≥10% at 16,384 BFF population updates (epochs)

e First / final inflow and fetch

Variant	Worlds n	H bits first / final	Fetch/256 first / final
Literal, no wrap	12	7.99 / 7.99	1.00 / 1.00
Standard	9	0.04 / 0.00	0.00 / 0.00
Pointer wrap	19	0.00 / 0.00	0.00 / 0.00
Wrap + literal	12	0.75 / 0.64	1.00 / 1.00
Wrap + literal, harmless	12	0.04 / 0.00	1.00 / 1.00

First = first-modal tape; final = terminal modal tape

Paired world sampling; values are medians across worlds
Fetch/256 counts any-partner-fetch encounters; H units as b,c

f Registered tests of partner inflow (H)

Per-world criterion H in bits	Passing worlds, n/N	Support / reject % passing	Verdict
Q1 Z80 first H>1	80/80	≥90% / ≤60%	met
Q2 Z80 loop final H<0.5 67 worlds: loop final tapes	63/67	≥90% / ≤60%	met
Q3 Z80 H_first − H_final ≥2	63/80	≥75% / ≤50%	met
Q4a BFF standard first H<0.5	6/9	≥90% / ≤60%	between
Q4b BFF pooled literal variants: first H>1	12/36	≥90% / ≤60%	rejected

Q4b: Literal, no wrap; Wrap + literal; Wrap + literal, harmless

12 worlds each; 36 total; same names as a, b and e
Q1–3: Z80 Stage G (80 worlds); Q2: 67 worlds with loop-bearing final tapes
Q4a: Standard BFF; Q4b: the three pooled variants named above
Support/reject use pass/world fraction; between = neither threshold